A COMMUTATIVE SQUARE OF SITES WITH CONTINUOUS AND COCONTINUOUS FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Sheafification is not needed for simultaneously continuous and cocontinuous functors. Same chapter, Lemma lemma-when-shriek. Removes the sheafification in the descriptions of the pullbacks.

Source reference. Original section 00X0: sources/stacks-sites/pinned-original-source.tex, source lines 6929–6929. Navigation pointer invoked at proof line 6929; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 00XN: sources/stacks-sites/pinned-original-source.tex, source lines 6930–6930. Navigation pointer invoked at proof line 6930; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XR: sources/stacks-sites/pinned-original-source.tex, source lines 6857–6857. Navigation pointer invoked at proof line 6857; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Assume given sites $\mathcal{C}', \mathcal{C}, \mathcal{D}', \mathcal{D}$ and functors*

$$\begin{array}{ccc} \mathcal{C}' & \xrightarrow{v'} & \mathcal{C} \\ {\scriptstyle u'}\uparrow & & \uparrow{\scriptstyle u} \\ \mathcal{D}' & \xrightarrow{v} & \mathcal{D} \end{array}$$

With notation as in Sections 00X0 and 00XN assume

(1) *u and u' are continuous giving rise to morphisms of sites f and f',*
(2) *v and v' are cocontinuous giving rise to morphisms of topoi g and g',*
(3) *$u \circ v = v' \circ u'$, and*
(4) *v and v' are continuous as well as cocontinuous.*

Then[1] *$f'_* \circ (g')^{-1} = g^{-1} \circ f_*$ and $g'_! \circ (f')^{-1} = f^{-1} \circ g_!$.*

Proof. Namely, we have

$$f'_*(g')^{-1}\mathcal{F} = (u')^p((v')^p\mathcal{F})^\# = (u')^p(v')^p\mathcal{F}$$

The first equality by definition and the second by Lemma 00XR. We have

$$g^{-1}f_*\mathcal{F} = (v^p u^p \mathcal{F})^\# = ((u')^p(v')^p\mathcal{F})^\# = (u')^p(v')^p\mathcal{F}$$

The first equality by definition, the second because $u \circ v = v' \circ u'$, the third because we already saw that $(u')^p(v')^p\mathcal{F}$ is a sheaf. This proves $f'_* \circ (g')^{-1} = g^{-1} \circ f_*$ and the equality $g'_! \circ (f')^{-1} = f^{-1} \circ g_!$ follows by uniqueness of left adjoints. □

[1] In this generality we don't know $f \circ g'$ is equal to $g \circ f'$ as morphisms of topoi (there is a canonical 2-arrow from the first to the second which may not be an isomorphism).